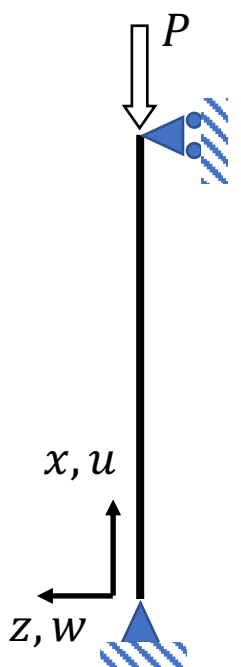


Equilibrium
 $\delta V = 0$

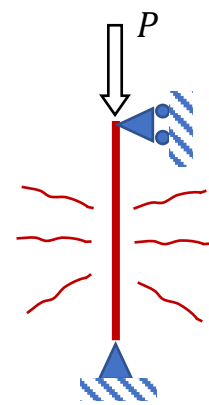
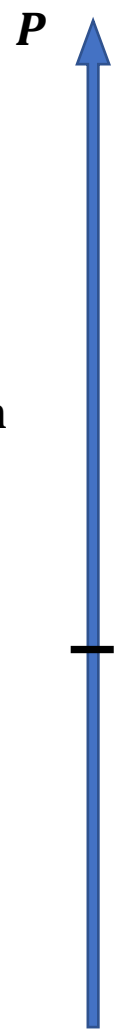


$$\varepsilon_{xx} = u_{,x} + \frac{1}{2} w_{,x}^2$$

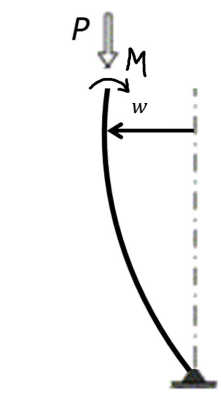
Geometric nonlinearity:
 axial strains $\longleftrightarrow$ beam rotation

Negative ε_{xx} $\rightarrow$ Reduced bending stiffness

Positive ε_{xx} $\rightarrow$ Increased bending stiffness



Unstable
 $\delta^2 V < 0$



Neutral Equilibrium Criterion

$\delta^2 V = 0$

Stable
 $\delta^2 V > 0$